

I. System jumper & connectors

X1 : 2xRS232

- 1 – RX input RS232 -> MAX -> RX STM32 PA10 - 43
- 2 – TX output RS232 <- MAX <- TX STM32 PA9 - 42
- 3 – GND
- 4 – RX input RS232 -> MAX -> & TX STM32 PC6 - 37 -> RX EP1C6 - 119; X13 - 2
- 5 – TX output RS232 <- MAX <- RX STM32 PC7 - 38 <- TX EP1C6 - 113; X13 - 3

X2 : RS485

- 1 – RS485 - A
- 2 – RS485 - B
- 3 – GND
- 4,5 – RS485 120 Ohm enable jumper

X3 : CAN

- 1 – CAN - L
- 2 – CAN - H

X4 : CAN 120 Ohm enable jumper

X5 : Microcontroller BOOT mode (1-2:PIN PC13 BOOT) or (2-3:BOOT0=1)

- 1 – PC13 pin
- 2 – BOOT0 pin
- 3 – VDD 3.3V

X6 : Power external input +5V

- 1 – GND
- 2 – +5V Board

X7 : USB Type-C Power Jumper

- 1 – +5V Type-C
- 2 – +5V Board

X8 : +5V output from USB Type-C

- 1 – +5V Type-C
- 2 – GND

X9 : Watchdog timer jumper (1-2:WDTON) or (2-3:WDTOFF)

- 1 – Reset WDT & Button
- 2 – Reset
- 3 – Reset Button

X10 : Processor BOOT mode enable jumper

X11 : Speaker Output

- 1 – OUT_M
- 2 – OUT_P

X12 : V3S VRTC (1-2:VRTC = VDD 3.3V) or (2-3:VRTC=GND)

1 – VDD 3.3V

2 – V3S VCC-RTC - 98

3 – GND

X13 : IN-OUT

1 – V3S - 100

2 – EP1C6 - 119

3 – EP1C6 - 113

X14 : BAT Power

1 - +V BAT

2 - GND

II. Pin Connectors

Y1 : STM32F412

01 - STM32F412[61] - PB8 / I2C1_SCL / TIMER4_CH3 / CAN1_RX

02 - STM32F412[40] - PC9 / MCO2

03 - STM32F412[62] - PB9 / I2C1_SDA / TIMER4_CH4 / CAN1_TX

04 - STM32F412[24] - PC4 / ADC1_IN14

05 - STM32F412[28] - PB2 / BOOT1

06 - STM32F412[22] - PA6 / ADC1_IN6 / TIMER3_CH1

07 - STM32F412[27] - PB1 / TIMER3_CH4

08 - STM32F412[21] - PA5 / ADC1_IN5 / TIMER2_CH1

09 - STM32F412[26] - PB0 / ADC1_IN8 / TIMER1_CH2N / TIMER3_CH3

10 - STM32F412[20] - PA4 / ADC1_IN4 / UART2_CK

11 - STM32F412[36] - PB15 / SPI2_MOSI / TIMER1_CH3N

12 - STM32F412[10] - PC2 / ADC1_IN12 / SPI2_MISO

13 - STM32F412[29] - PB10/ SPI2_SCK / TIMER2_CH3

14 - STM32F412[09] - PC1 / ADC1_IN11

15 - VDD_3.3V

16 - STM32F412[08] - PC0 / ADC1_IN10

17 - GND

18 - STM32F412[15] - PA1 / ADC1_IN1 / TIMER2_CH2 / TIMER5_CH2

19 - STM32F412[35] - PB14/ SPI2_MISO / TIMER1_CH2N

20 - STM32F412[14] - PA0 / ADC1_IN0 / TIMER2_CH1 / TIMER5_CH1

Y2 : SYSTEM

01 - V3S[49] - PB9 / UART0_RX - Y4[44]

02 - GD32F103[21] - PA09/ USART0_TX

03 - V3S[48] - PB8 / UART0_TX - Y4[42]

04 - GD32F103[22] - PA10/ USART0_RX

05 - VDD_3.3V

06 - GD32F103[25] - PA13 / SWDIO

07 - NC

08 - GD32F103[28] - PA14 / SWCLK

09 - RESET

10 - GND

Y3 : CMSYS-DAP

- 01 - VDDIN
- 02 - GD32F103[13] -PA6 / nRST control
- 03 - GD32F103[12] -PA5 / TDO control
- 04 - GD32F103[11] -PA4 / TDI control
- 05 - GD32F103[08] -PA1 / SWDIO control
- 06 - STM32F412[46] -PA13/ SWDIO in
- 07 - GD32F103[07] -PA0 / SWCLK control
- 08 - STM32F412[49] -PA14/ SWCLK in
- 09 - VDD_3.3V
- 10 - GND

Y4 : COMMON

- 01 - GND
- 02 - VDD_5.0V
- 03 - V3S[15] - LCD_D18
- 04 - V3S[14] - LCD_D19
- 05 - V3S[13] - LCD_D20
- 06 - V3S[11] - LCD_D21
- 07 - V3S[07] - LCD_D22
- 08 - V3S[06] - LCD_D23
- 09 - V3S[24] - LCD_D10
- 10 - V3S[23] - LCD_D11
- 11 - V3S[22] - LCD_D12
- 12 - V3S[18] - LCD_D13
- 13 - V3S[17] - LCD_D14
- 14 - V3S[16] - LCD_D15
- 15 - V3S[33] - LCD_D2
- 16 - V3S[32] - LCD_D3
- 17 - V3S[31] - LCD_D4
- 18 - V3S[30] - LCD_D5
- 19 - V3S[28] - LCD_D6
- 20 - V3S[27] - LCD_D7
- 21 - V3S[37] - LCD_CLK
- 22 - V3S[35] - LCD_HS
- 23 - V3S[34] - LCD_VS
- 24 - V3S[36] - LCD_DE
- 25 - EP1C6[129] - IO.00
- 26 - V3S[39] - PB0/UART2_TX
- 27 - EP1C6[130] - IO.01
- 28 - V3S[40] - PB1/UART2_RX
- 29 - EP1C6[131] - IO.02
- 30 - V3S[41] - PB2/UART2_RTS
- 31 - EP1C6[132] - IO.03
- 32 - V3S[42] - PB3/UART2_CTS
- 33 - EP1C6[142] - IO.04
- 34 - V3S[43] - PB4/PWM0
- 35 - EP1C6[143] - IO.05
- 36 - V3S[43] - PB5/PWM1
- 37 - EP1C6[144] - IO.06
- 38 - V3S[45] - PB6/TWI0_SCL
- 39 - EP1C6[01] - IO.07
- 40 - V3S[46] - PB6/TWI0_SDA
- 41 - EP1C6[02] - IO.08
- 42 - V3S[48] - PB8/UART0_TX

43 - EP1C6[03]	- IO.09
44 - V3S[49]	- PB9/UART0_RX
45 - EP1C6[04]	- IO.10
46 - V3S[09]	- PE21/TWI1_SCL
47 - EP1C6[05]	- IO.11
48 - V3S[08]	- PE22/TWI1_SDA
49 - ADS1248[06]	- REFNO
50 - ADS1248[05]	- REFPO
51 - ADS1248-R	- ERTDG
52 - ADS1248-R	- ERTD
53 - ADS1248[11]	- IN0
54 - VDDIN	
55 - ADS1248[13]	- IN4
56 - ADS1248[12]	- IN1
57 - ADS1248[15]	- IN6
58 - ADS1248[14]	- IN5
59 - ADS1248[20]	- IEXC1
60 - ADS1248[16]	- IN7
61 - ADS1248[17]	- IN2
62 - ADS1248[19]	- IEXC2
63 - ADS1248[18]	- IN3
64 - AGND	

Y5 : JTAG

01 - JTAG_TCK	- EP1C6[88]
02 - GND	
03 - JTAG_TDO	- EP1C6[90]
04 - nCONFIG	- EP1C6[14]
05 - JTAG_TMS	- EP1C6[89]
06- CONFDONE	- EP1C6[86]
07- STATUS	- EP1C6[87] - GD32F103[16]
08 - DATA	- EP1C6[13] - GD32F103[34]
09 - JTAG_TDI	- EP1C6[95]
10 – DCLK	- EP1C6[24] - GD32F103[33]

Y6 : HUB-75

01 - EP1C6[99]	- IO.34
02 - EP1C6[100]	- IO.35
03 - EP1C6[96]	- IO.33
04 - GND	
05 - EP1C6[97]	- IO.32
06 - EP1C6[85]	- IO.31
07 - EP1C6[84]	- IO.30
08 - EP1C6[83]	- IO.29
09 - EP1C6[82]	- IO.28
10 - EP1C6[79]	- IO.27
11 - EP1C6[78]	- IO.26
12 - EP1C6[77]	- IO.25 - LED WHITE
13 - EP1C6[76]	- IO.24
14 - EP1C6[75]	- IO.23
15 - EP1C6[74]	- IO.22
16 - GND	

Y7 : ANALOG

- 01 - ADCA - U34_ADS7884[3] Input
- 02 - ADCB - U35_ADS7884[3] Input
- 03 - DACA - U30_DAC8830 - OP.U21[1] Output
- 04 - DACB - U27_DAC8830 - OP.U21[7] Output
- 05 - AGND

III. Board Connectors

Z1 : micro USB

- 01 - VBUS
- 02 - DM - STM32F412[44] PA11-USBDM
- 03 - DP - STM32F412[45] PA12-USBDP
- 04 - ID - GND
- 05 - GND

Z2 : USB Type-C

- 01 - GND
- 02 - VBUS
- 03 - NC
- 04 - NC
- 05 - DP - GD32F103[24] PA12-USBDP
- 06 - DM - GD32F103[23] PA11-USBDM
- 07 - DP - GD32F103[24] PA12-USBDP
- 08 - DM - GD32F103[23] PA11-USBDM
- 09 - NC
- 10 - NC
- 11 - VBUS
- 12 - GND

Z3 : AUDIO JACK

- 01 - MIC
- 02 - HPCOM
- 03 - HPR
- 04 - HPL

Z4 : USB-A

- 01 - VDD_5.0V
- 02 - USB-DM - V3S[110]
- 03 - USB-DP - V3S[111]
- 04 - GND

Z5 : micro SD

- 01 - SDC0_DAT2 - V3S[101]
- 02 - SDC0_DAT3 - V3S[102]
- 03 - SDC0_CMD - V3S[103]
- 04 - VDD_3.3V
- 05 - SDC0_SCK - V3S[105]
- 06 - GND
- 07 - SDC0_DAT0 - V3S[106]
- 08 - SDC0_DAT1 - V3S[107]
- 09 - NC